

Letter

Structural characterization, electrical and magnetic properties of $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ filmsCunquan Li¹, Jiashuo Yan¹, Yankai Chen, Yukai An^{*}

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ABSTRACT

Structural, electrical, and magnetic properties of $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films with different Fe concentration are investigated. The doped Fe ions replace the In1 sites of In_2O_3 lattice with the mixed-valence of +2 and +3 states and form $\text{Fe}_{\text{In}1}+2\text{V}_\text{O}$ complex based on the real-space multiple-scattering simulation. The room temperature ferromagnetic hysteresis loop of Fe/Mg co-doped In_2O_3 films shows ferromagnetic ordering with coercivity H_c of about 150–300 Oe. Obvious increase of ferromagnetic hysteresis loop and saturation magnetic moment with a maximum value of 8.45 emu/cm^3 ($x = 0.08$) are observed due to the more incorporation of Fe ions. All the films exhibit n-type conductive behavior, which are mainly dominated by Mott variable range hopping with strong localization of carriers. Interestingly, the contribution of positive and negative magnetoresistance (MR) strongly correlates with Fe dopant and temperature. It can conclude that the $\text{Fe}_{\text{In}1}+2\text{V}_\text{O}$ complex based on bound magnetic polaron model plays a key role for achieving the intrinsic ferromagnetism.

1. Introduction

Recently, dilute magnetic semiconductors (DMSs) formed by replacing the non-magnetic cations of semiconductor with transition metal (TM) attract wide interest due to possess the magnetic and semiconductor properties. The DMSs materials can utilize the electron spin and charge properties to process, transfer, and store data in spintronic devices. Since Dietl et al. theoretically predicted that TM-doped p-type ZnO can exhibit the ferromagnetic ordering with above room-temperature (RT) [1], metal oxide semiconductor gradually become hot topic due to the large possibility of controlling both the spin and charge. Recently, high temperature ferromagnetic ordering has been observed in the TM-doped ZnO [2–6], TiO_2 [7–10], SnO_2 [11–13] and In_2O_3 [14–18] systems. Among these oxides, In_2O_3 with wide bandgap of 3.75 eV and excellent properties receives much interest and has become a promising DMS material [15]. Many research teams have investigated the spin-related ferromagnetic properties of In_2O_3 based DMSs for the applications of spintronic devices. He et al. first proved the appearance of intrinsic ferromagnetic ordering in the Fe doped In_2O_3 [19]. Park et al. found that the maximum saturation magnetization (Ms) can reach to 7.1 emu/cm^3 in the Mo doped In_2O_3 films, which is attributed to carrier-mediated direct exchange interactions [20]. Jiang

et al. concluded that the large RT ferromagnetic ordering in Fe/Sn co-doped In_2O_3 powders is extrinsic, which originates from the contribution of Fe_3O_4 impurity [21]. Dhamodaran et al. observed that Fe, Mn co-doping can lead to significant intrinsic RT ferromagnetism in In_2O_3 , and the increase of defect concentration can control and improve the ferromagnetism [22]. Jiang et al. deposited the In_2O_3 films co-doped with Fe/Cu atoms and observed the inhomogeneous RT ferromagnetism [23]. Wang et al. proved that the oxygen vacancy defects play an important role on the ferromagnetic interaction in doped In_2O_3 systems [24]. Liu et al. deposited the Mn and Mg co-doped In_2O_3 films, which exhibited a non-monotonous intrinsic RT ferromagnetic behavior [25]. The observed ferromagnetism in doped In_2O_3 systems possibly originates from the magnetic impurity phases, which result an extrinsic magnetic mechanism. On the other hand, some intrinsic ferromagnetic mechanisms have been suggested, including super-exchange magnetic interaction between magnetic ions, carrier-mediated magnetic interaction and bound magnetic polaron mechanism based on structural defects (such as oxygen vacancy), etc. Therefore, to clarify the magnetic origin or possible magnetic interactions in the doped or co-doped In_2O_3 systems, it is important to understand how the magnetic dopants adjust and alter the magnetic and transport properties of In_2O_3 based DMSs, in accompany with the change of structural defects. In this paper, the

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Mg/Fe co-doped In_2O_3 films with different Fe concentrations are prepared by magnetron sputtering technology. The extend X-ray absorption fine structure (EXAFS) and near edge X-ray absorption fine structure (NEXAFS) at Fe K-edge are measured to determine the local structures (occupation site, metallic cluster, and oxide, et. al.) of doped Fe atoms. The corresponding physical properties, such as carrier concentration, ρ -T curves, hysteresis loop and magnetoresistance (MR) of Mg/Fe co-doped In_2O_3 films are investigated in detail, which helps to clarify the magnetic mechanisms and the corresponding correlation with the structural defect, Fe dopant and carrier.

2. Experimental details

A series of $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films are deposited using RF-magnetron sputtering. The vacuum level before deposition is 8.7×10^{-5} Pa and the temperature of substrate is kept at 450°C . During the process of deposition, the sputtering pressure is 0.8 Pa and the Ar gas flow rate is 10 sccm. The $(\text{In}_{0.97}\text{Mg}_{0.03})_2\text{O}_3$ ceramic target with some high-purity Fe chips symmetrically attached is exploited to prepared the $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films. By altering the number of Fe chips, the $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films with $x = 0, 0.02, 0.05, 0.08$, and 0.11 are obtained. The concentrations of Mg and Fe atoms are estimated using the energy dispersive X-Ray spectroscopy. The thickness of film is about 500 nm, which is decided by JOEL-6700F scanning electron microscope. The crystal structure is detected by Rigaku Dmax2500PC X-ray diffraction with $\lambda = 0.15406$ nm. The Fe K-edge XANES and EXAFS spectra are measured at the diffraction station (4B9A) of Beijing Synchrotron Radiation Facility. The hysteresis loop of films is carried out on the PPMS-9 system. The Hall effect measurement is used to estimate the carrier concentration, conductive type and resistivity by van der Pauw method. The measurements of ρ -T and M-R curves are carried out by the physical properties measurement system (PPMS-9 Quantum Design) using the four-probe method.

3. Results and discussion

3.1. Structural properties of $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films

Fig. 1 displays the XRD images of $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ ($x = 0.02, 0.05, 0.08, 0.11$) films. These peaks well match the bixbyite structures (space group $\text{Ia}\bar{3}$) of In_2O_3 (ICDD card No. 71-2194) and SiO_2 (ICDD card No. 84-0791), and the second phases of Mg/Fe and their oxides are not observed in the films. One can see from the inset of Fig. 1, the (222) peak shifts to higher angle with increasing Fe doping concentration, indicating a decreasing lattice constant (Bragg formula: $2d\sin\theta = n\lambda$) [26]. The radius of Fe^{3+} and Fe^{2+} ions are 0.79 and 0.92 Å, respectively

[27], which are smaller than the radius of In^{3+} ions. Therefore, the gradual decrease of lattice constant a with Fe doping implies that the doped Fe ions substitute for the In^{3+} sites into the In_2O_3 lattice. Considering the accuracy of XRD measurement, even if no second magnetic phases are observed in the XRD patterns, the possibility of the existence of impurity phases in the films cannot be excluded, which requires further supplement by other detection method.

Fig. 2(a) displays the Fe K-edge enlarged XANES spectra for the $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films and standard samples of Fe, FeO, Fe_2O_3 and Fe_3O_4 . Clearly, there exists a distinct edge-front peak around 7113 eV for the films, which originates from the 1s-3d transition due to the Fe 3d/O 2p orbital hybridization [28]. The absorption edge of XANES spectra for all the films is higher than that of FeO reference and is lower than that of Fe_2O_3 reference, suggesting that the doped Fe ions in the films mainly exists in the form of +2 and +3 mixed valence states. The XANES spectrum is a good fingerprint for the oxygen vacancy (V_O) and In interstitial (In_i) defects around Fe dopant in the $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films. The Fe K-edge XANES spectra can be well simulated using the real-space multiple-scattering approach with FEFF 9.0 code. Fig. 2(b) and (c) show the experimental and calculated XANES spectra with M_1 - M_7 model structures. The model M_1 : Fe replaces In_1 with In interstitial around it; Model M_2 : Fe replaces In_2 with In interstitial around it; Model M_3 : Fe replaces In_2 with one oxygen vacancy around it; Model M_4 : Fe replaces In_2 ; Model M_5 : Fe replaces In_1 with two oxygen vacancies around it; Model M_6 : Fe replaces In_1 with one oxygen vacancies around it; Model M_7 : Fe replaces In_1 . It can be clearly seen that there exists great difference between the calculated XANES spectra of M_1 ($\text{Fe}_{\text{In}1} + \text{In}_i$), M_2 ($\text{Fe}_{\text{In}2} + \text{In}_i$), M_3 ($\text{Fe}_{\text{In}2} + V_O$) and M_4 ($\text{Fe}_{\text{In}2}$). However, the calculated XANES spectra of M_5 ($\text{Fe}_{\text{In}1} + 2V_O$), M_6 ($\text{Fe}_{\text{In}1} + V_O$) and M_7 ($\text{Fe}_{\text{In}1}$) exhibit similar characteristics as the experimental spectrum. One can see that the model structure M_5 , namely the doped Fe ion replaces In_1 atom with two oxygen vacancies in the nearest coordination shell well simulates the characteristics of experimental XANES spectrum. This is also a powerful witness for the existence of oxygen vacancies around the Fe dopant in $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films. Fig. 2(d) displays the Fourier transforms (FT) of EXAFS $k^3\chi(k)$ functions measured at Fe K-edge for the $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films and standard samples as well as the Fourier transform at In K-edge for the pure In_2O_3 with the fitting curve using dashed lines. The FT curves for all the films are basically consistent. The peaks of A, B, and C show the Fe-O first-nearest neighbor, Fe-In second-nearest neighbor and Fe-In third-nearest neighbor, with corresponding distances of 1.52, 2.65, and 3.22 Å, respectively. The three peak positions are different from the nearest neighbor coordination of standard samples, but are very consistent with the coordination structure of pure In_2O_3 [15]. This further proves that the Fe element in the films is substitutional, rather than in the form of Fe metal and its oxide. To study the coordination, bond length, and other

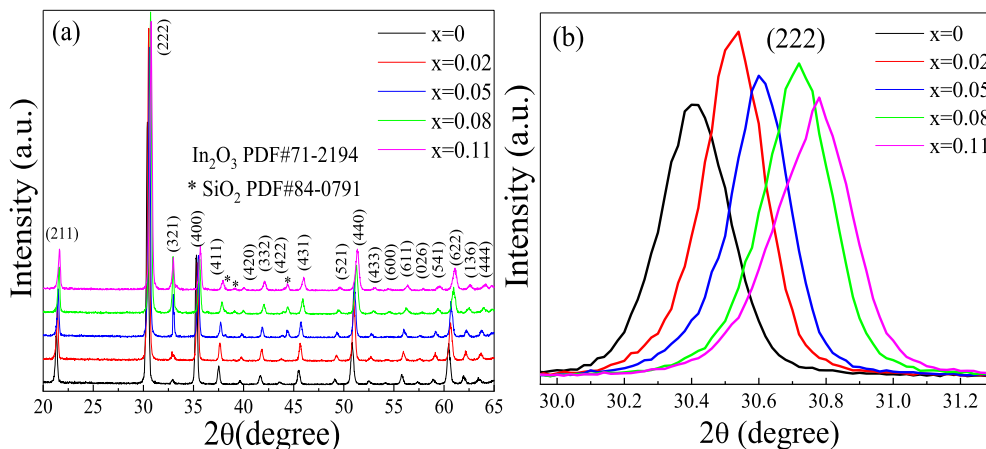


Fig. 1. (a) XRD images and (b) magnified images of (222) peaks for the $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ ($x = 0.02, 0.05, 0.08, 0.11$) films.

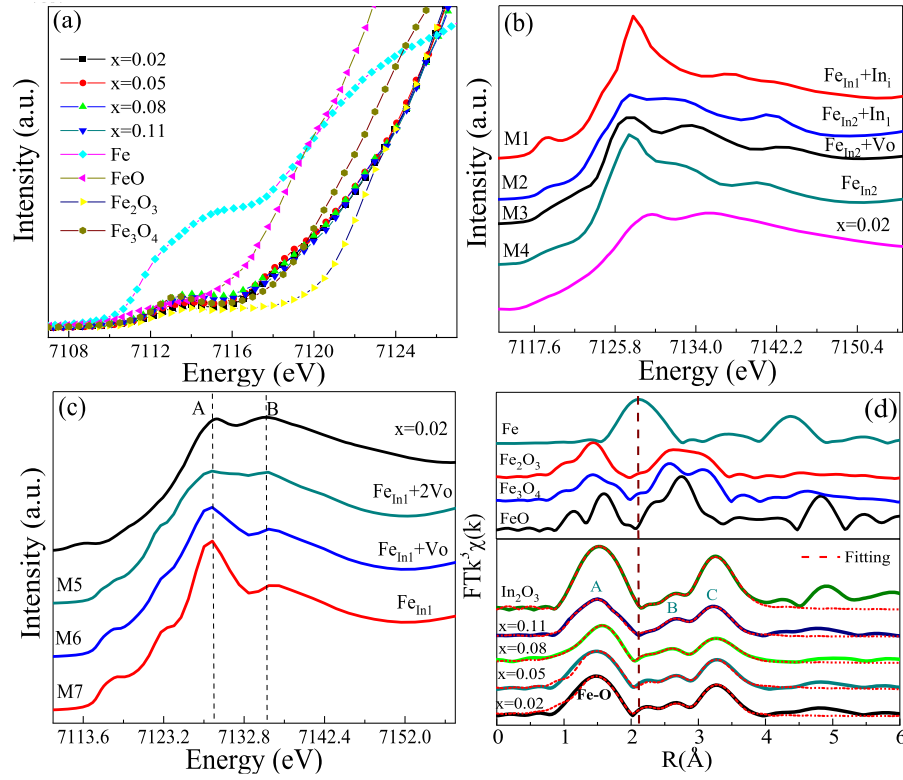


Fig. 2. (a) Fe K-edge enlarged XANES spectra for the $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films and standard samples of Fe, FeO, Fe_2O_3 and Fe_3O_4 . (b) Experimental and calculated XANES spectra with M1–M4 model structures. (c) Experimental and calculated XANES spectra with M5–M7 model structures. (d) Fourier transforms of EXAFS $k^3\chi(k)$ functions measured at Fe K-edge for the $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films and standard samples of Fe, FeO, Fe_2O_3 and Fe_3O_4 as well as the Fourier transform at In K-edge for the pure In_2O_3 with the fitting curve using dashed lines.

local environments of doped Fe ions, we used Artemis software to fit the FT curves of films with different Fe doping concentrations. The fitted curve is displayed using the dashed line of Fig. 2(d). The fitting process adopts the model of Fe substitution for In_1 site with a fitting range of 0.4 Å to 4.0 Å. The corresponding fitted parameters are displayed in Table 1. Since the radius of $\text{Fe}^{2+}/\text{Fe}^{3+}$ ions are smaller than that of In^{3+} ion, the Fe substitution for In^{3+} will cause lattice compression, which will reduce the bond length accordingly. Therefore, the bond lengths of Fe–O, Fe– In_1 and Fe– In_2 neighbors are slightly smaller than those in pure In_2O_3 (2.20 Å, 3.37 Å and 3.83 Å). With Fe doping, the contraction effect of lattice increases, and the corresponding bond length also decreases. At the same time, the Debye-waller factor σ^2 obviously increases and the effective coordination numbers around three neighboring coordination shells of Fe ion decrease. Therefore, after the doped Fe ions enter the In_2O_3 lattice, it will cause the distortion of local lattice and Fe–O bonds, resulting in an increase of lattice disorder. However, the disorder factor σ^2 for the second Fe– In_1 and the third Fe– In_1 nearest neighbors is less affected.

3.2. Electrical and magnetic properties of $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films

Fig. 3(a)–(c) shows the ρ –T curves of $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films. Resistivity of film with $x = 0.11$ at low temperature is so high, which is beyond the measurement range. Therefore, only the ρ –T curves with $x = 0.02$, 0.05 and 0.08 are displayed. At around 191 K, there exists a transition of conductive from semiconductor to metallic characters for the film with $x = 0.02$, which is also observed in Ni-doped ITO bulk samples [29]. On the contrary, the films with higher Fe concentration exhibits semiconductor conductivity characteristics in the temperature range from 10 K to 300 K. Generally, the resistivity of impurity semiconductors correlates with the carrier concentration, mobility, and temperature. There are two scattering mechanisms in impurity semiconductors: ionized impurity scattering and lattice scattering. So, impurity ionization and intrinsic excitation become the main sources of resistivity of impurity semiconductor. To understand the carrier transport mechanism of $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films, the corresponding $\ln\rho \sim T^{-1/4}$ curves are fitted by applying $\rho = A\exp[(T_0/T)^{1/4}] + B\exp(E_d/k_B T)$ formula, are shown in the solid line of Fig. 3(d). Where the $A\exp[(T_0/T)^{1/4}]$ is the contribution of Mott variable range hopping to the

Table 1

Fitted parameters of Fourier transforms of EXAFS $k^3\chi(k)$ functions measured at Fe K-edge. N , R and σ^2 are the coordination number, bond length and e Debye-waller factor, respectively.

Samples x	Structural model for substitutional Fe								
	Fe–O			Fe– In_1			Fe– In_2		
	N_1	$\sigma_1^2(\text{\AA}^2)$	$R_1(\text{\AA})$	N_2	$\sigma_2^2(\text{\AA}^2)$	$R_2(\text{\AA})$	N_3	$\sigma_3^2(\text{\AA}^2)$	$R_3(\text{\AA})$
0.02	5.21	0.00538	2.068	5.21	0.00342	3.324	5.21	0.00338	3.826
0.05	5.04	0.00617	2.064	5.04	0.00375	3.323	5.04	0.00347	3.822
0.08	4.78	0.00685	2.061	4.78	0.00390	3.325	4.78	0.00357	3.821
0.11	4.65	0.00736	2.057	4.65	0.00404	3.321	4.65	0.00364	3.817

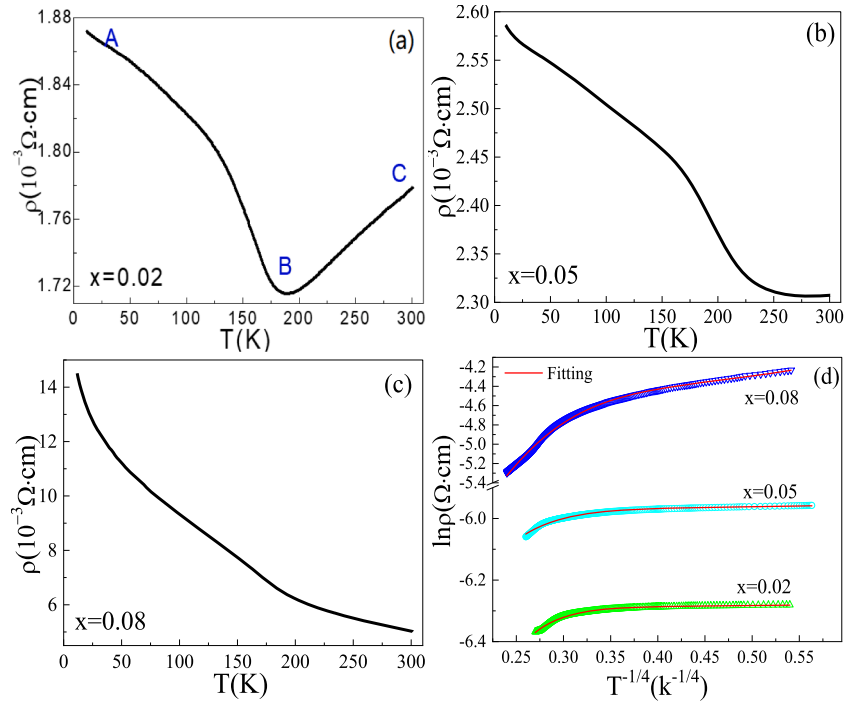


Fig. 3. ρ - T curves of $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films with $x =$ (a) 0.02, (b) 0.05 and (c) 0.08. (d) The $\ln\rho$ - $T^{-1/4}$ and the corresponding fitting curves.

resistivity with neglectable the Coulomb interaction between carriers. $\text{Bexp}(E_d/k_B T)$ is the contribution of hard band gap hopping to the resistivity with a constant band gap in the Fermi level. Clearly, the fitting results are perfect, indicating that the transport mechanisms are dominated by Mott variable range hopping at low-temperature region and hard band gap hopping at high-temperature region. Based on the measured results of Hall measurements and the fitted parameters, the relevant parameters of transport properties of films can be calculated using formulas:

$$K_F l = \frac{\hbar(3\pi^2)^{2/3}}{(e^2 \rho n^{1/3})} \quad (1)$$

$$\xi = (2\beta/N_d)^{1/3} (T_v/T_0)^{1/4} \quad (2)$$

where ρ is resistivity, n is carrier concentration, l is mean free path, T_0 is the characteristic hopping temperature, ξ is the localization radius, and T_v is onset temperature of hopping of ρ - T curve. The corresponding fitting and calculated parameters are displayed in Table 2. The Hall measurements suggest that all the films are n-type semiconductors. The substitutional Fe^{2+} ions for the In^{3+} ions provide holes, which compensate for the electrons provided by the O vacancies in the In_2O_3 lattice itself. Thus, the carrier concentration n of films decreases and ρ increases ($\rho \propto n^{-1}$) with increasing the Fe doping concentration. The Fe doping concentration is relatively low, and the holes provided cannot fully compensate for the electrons. Thus, all the films still maintain its original n-type conductivity. The $K_F l$ values of $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films are all less than 1, indicating strong localization of charge carriers. As the Fe concentration increases, the scattering effect of electrons

becomes greater, and the average free path also gradually decreases. Due to the hopping probability of carriers is inversely proportional to T_0 , thus the hopping probability of carriers decreases with the increase of Fe concentration. On the other hand, it can consider that the Fe doping can increase the localization effect of carriers, further resulting in a decrease in hopping localization radius of carrier.

Fig. 4(a) displays the hysteresis loops of $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films measured at 300 K. The diamagnetic signals contributed by the Si substrates have been removed. Clearly, all the films exhibit room temperature ferromagnetism. The open and symmetric hysteric behavior with coercivity H_c of about 150-300 Oe at lower fields can be observed, as shown in the inset of Fig. 4(a). As can be seen from Fig. 4(b), as the Fe concentration (x) increases, the M_s sharply increases to a maximum value of 8.45 emu/cm³ with $x = 0.08$ and then decreases. According to the XANES and EXAFS analysis, the doped Fe ions substitute for the In^{3+} sites of In_2O_3 lattice with two oxygen vacancies in the nearest coordination shell for the $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films. Therefore, the observed ferromagnetism in the films does not originate from the impurity magnetic phases, and should be intrinsic. The measurements of transport properties further prove strong localization characters of carriers, implying that the magnetic mechanism does not originate from carriers. Generally, the highly non-equilibrium sputtering process makes it possible that there exists a great deal of V_O defects in the doped In_2O_3 films, which has been proved by the XANES results. Therefore, the In^{3+} - V_O - Fe^{2+} or Fe^{3+} - V_O - Fe^{2+} structures are easily formed, and the $\text{Fe}_{\text{In}1}+2\text{V}_\text{O}$ complex based on bound magnetic polaron (BMP) model plays a key role for achieving the observed intrinsic ferromagnetism [30,31]. In BMP, the V_O forms F-centres with trapped electrons, which occupy a like hydrogen orbital with a polarization radius of $r_{\text{BMP}} = m k a_0 / m^*$. Within

Table 2

The resistivity ρ , carrier concentration n , mean free path l , characteristic hopping temperature T_0 , localization radius ξ , and onset temperature of hopping (T_v) for the $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films.

x	ρ ($10^{-3}\text{ohm}\cdot\text{cm}$)	n (10^{19}cm^{-3})	$K_F l$	l (nm)	T_v (K)	T_0 (K)	ξ (nm)
0.02	2.1	8.3	0.99	0.90	10.8	16842	3.80
0.05	3.2	6.5	0.32	0.41	13.7	31025	2.86
0.08	5.6	4.1	0.21	0.22	15.3	40126	2.61

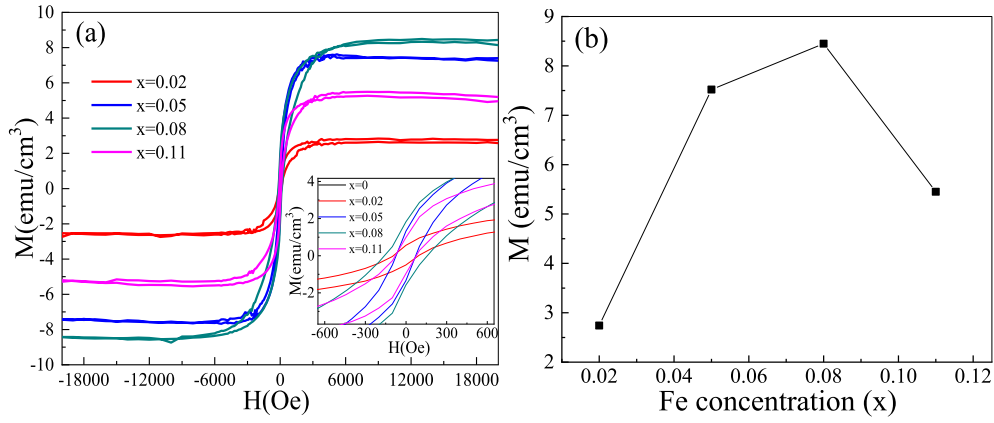


Fig. 4. (a) Hysteresis loops for the $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films measured at 300 K. The inset shows the enlarged loops at low magnetic field region. (b) The variation of M_s with Fe concentration (x).

the range of polaron radius, the spin of magnetic ions keeps the same direction, resulting in the local ferromagnetic order and the generation of BMP. The adjacent BMPs overlap with each other, forming the related clusters of BMPs that exhibit the ferromagnetism at the macro level. For our $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films, the M_s gradually increases to a maximum value of 8.45 emu/cm^3 and then decreases with Fe doping, indicating a nonmonotonic magnetic behavior. Possibly, with increasing the Fe doping concentration, the $\text{In}^{3+}\text{-Vo-Fe}^{2+}$ or $\text{Fe}^{3+}\text{-Vo-Fe}^{2+}$ structures in the films gradually increase, and the number of BMPs that can undergo overlap also increases, which is conducive to the occurrence of overlap between the BMPs. Finally, the M_s of films increases with Fe doping. When the Fe concentration continues to increase to $x = 0.11$, the high Fe doping concentration can lead more Fe ions outside the

ferromagnetic domains (r_{BMP}) and form $\text{Fe-O}^{2-}\text{-Fe}$ antiferromagnetic super-exchange interaction, which weakens the F-centre mediated ferromagnetic coupling and result in the decrease in M_s .

3.3. Magnetoresistance characteristics of $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films

Fig. 5(a)–(c) shows the variation of magnetoresistance (MR) with respect to magnetic field for the $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films under various temperature. Here, magnetoresistance is defined as $\text{MR} = \rho_H / \rho_0$. When $x = 0.02$, at a temperature of 10 K, the film shows a trend from negative MR to positive MR around 1 T. However, as the temperature increases, the MR gradually increases and exhibit the negative MR behavior. When $x = 0.05$, at a temperature of 10 K, the film undergoes

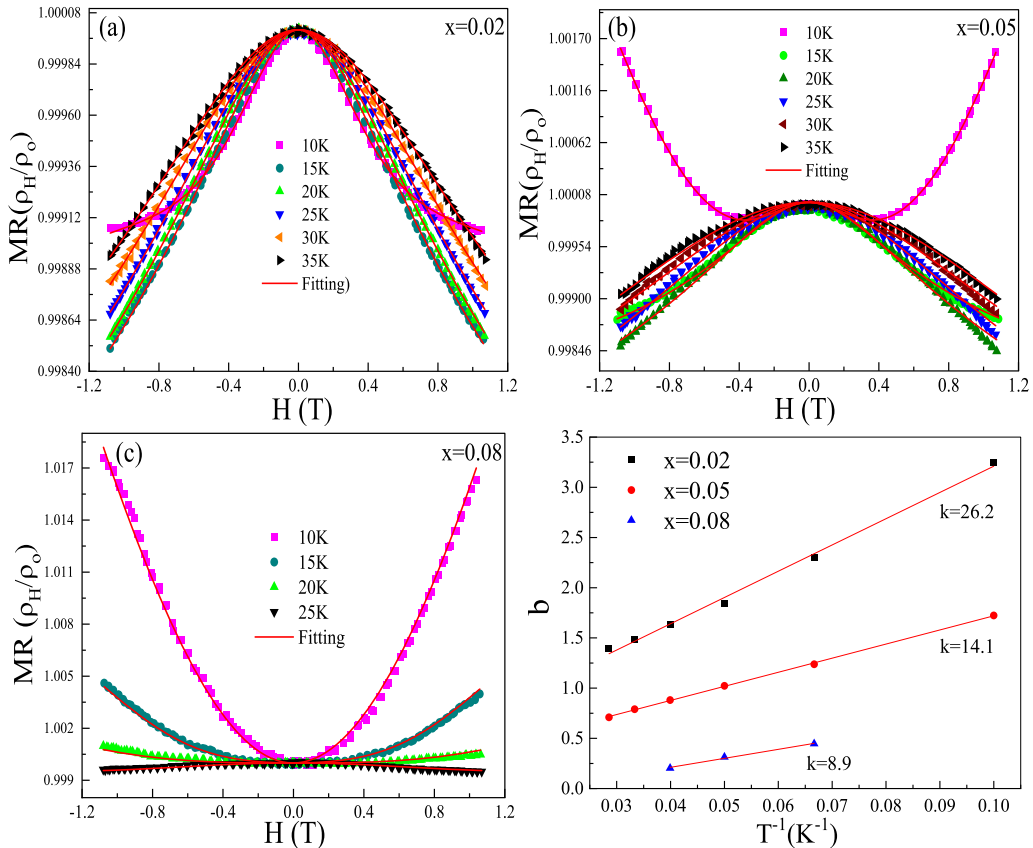


Fig. 5. Variation of MR with respect to magnetic field for the $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films with $x =$ (a) 0.02, (b) 0.05 and (c) 0.08 under various temperature. (d) The fitted parameter b vs inverse temperature under various Fe concentration.

an obvious transition from negative MR to positive MR around 0.38 T. When the temperature rises to 15 K, the positive MR behavior around 1 T is observed. After the temperature continues to rise, the negative MR character dominates throughout the range of -1.08 T to 1.08 T. When $x = 0.08$, the film exhibits obvious positive MR character at the low temperature, and as the temperature increases, the positive MR remarkably decreases. When the temperature reaches to 25 K, the negative MR occurs. In order to better study the MR characters of $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films, semi empirical formulas are applied to analyze the MR behaviors [32–35].

$$\text{NegativeMR} : \frac{\rho_H}{\rho_0} = 1 - a^2 \ln(1 + b^2 H^2) \quad (3)$$

$$\text{PositiveMR} : \frac{\rho_H}{\rho_0} = 1 + \frac{c^2 H^2}{1 + d^2 H^2} \quad (4)$$

$$a^2 = A_1 J D(\epsilon_F) [s(s+1) + \langle M^2 \rangle] \quad (5)$$

$$b^2 = \left[1 + 4s^2 \pi^2 \left(\frac{2JD(\epsilon_F)}{g} \right)^4 \right] \cdot \left(\frac{g\mu_B}{\alpha k_B T} \right)^2 \quad (6)$$

$$c^2 = \frac{\sigma_1 \sigma_2 (\mu_1 + \mu_2)^2}{(\sigma_1 + \sigma_2)^2} \quad (7)$$

$$d^2 = \frac{(\sigma_1 \mu_2 - \sigma_2 \mu_1)^2}{(\sigma_1 + \sigma_2)^2} \quad (8)$$

where $D(\epsilon_F)$ is the state density around Fermi level, $\langle M \rangle$ is the average magnetic moment, J is magnetic exchange parameter, μ_B is Bohr magneton, and g is effective Lande factor of local magnetic moment, α is constant, σ is the conductivity, and μ is the mobility of carrier. When $x = 0.02$, the experimental MR curve at the temperature of 10 K can be well fitted using a combination of positive and negative MR and only negative MR behavior at higher temperatures are fitted. The similar methods are used for the films with $x = 0.05$ and 0.08 . The corresponding fitting parameters are displayed in Table 3. Obviously, the fitting results of MR curves for all the films under various temperatures are very consistent with the experimental curves, indicating that the selection of fitting parameters and models is reasonable. On the other hand, the fitted parameter a does not change significantly under various temperatures for the same film, but the parameter b gradually decreases with increasing temperature.

In order to better study the variation pattern and significance of parameter b , the b - T^{-1} curves are plotted and shown in Fig. 5(d). Clearly, there exists a good linear relation between the parameter b and T^{-1} for the $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films under various Fe concentrations, indicating that all fitting models used in these experimental curves are correct. Interestingly, the slope shows a gradual decrease trend with Fe doping, which clearly suggests that the local magnetic moment spin s , exchange interaction integral J and Fermi level state density in the film decrease. Due to the p-d or s-d exchange effect in the Fe/Mg co-doped In_2O_3 systems, each carrier is surrounded by a polarized magnetic ion electron cloud. When the external magnetic field acts on the film, the spin of doped magnetic ions will be arranged in parallel with magnetic field. So, the migration of carriers will be easier, which makes the decrease of resistivity with the increase of magnetic field, namely the negative MR character. When the magnetic field is larger, the wave function of carriers shrinks, the overlap is reduced, and the probability of electron transition between two states decreases. The resistivity increases with the increase of external magnetic field, exhibiting the positive MR character.

4. Conclusions

In summary, the structural, electrical, and magnetic properties of

Table 3

The fitting parameters of MR curves for the $(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films under various temperatures.

x	T(K)	a	b	c	d
0.02	10	0.02387	3.24811	0.025	0.44721
	15	0.02724	2.30114		
	20	0.0300	1.83931		
	25	0.03033	1.6351		
	30	0.03066	1.47919		
	35	0.0300	1.39435		
0.05	10	0.0968	1.72348	0.12985	0.65266
	15	0.12021	1.23737	0.12381	0.64343
	20	0.04301	1.02273		
	25	0.04171	0.88068		
	30	0.03975	0.78914		
	35	0.04583	0.70866		
0.08	10			0.56725	0.3352
	15	1.04735	0.44487	0.542	0.34441
	20	0.97175	0.31229	0.38264	0.29587
	25	1.1104	0.20219	0.27251	0.26687

$(\text{In}_{0.97-x}\text{Fe}_x\text{Mg}_{0.03})_2\text{O}_3$ films are investigated using XRD, NEXAFS, EXAFS, Hall effect, ρ -T and MR measurements. The doped Fe atoms replace for the In_1 sites of In_2O_3 lattices, accompanying with a large number of O vacancies around them. All the films with strong localization of charge carriers exhibit n-type conductive behaviors, which are mainly dominated by the Mott variable range hopping. The Fe dopant and temperature can effectively alter the MR characters, which undergoes an obvious transition from negative MR to positive MR. All the films exhibit the RT ferromagnetic ordering, which strongly correlates with the $\text{In}^{3+}\text{-V}_\text{O}\text{-Fe}^{2+}$ or $\text{Fe}^{3+}\text{-V}_\text{O}\text{-Fe}^{2+}$ structures formed by the O vacancies around Fe dopant. It can conclude that the O vacancy defects are the key factor for achieving the RT ferromagnetic ordering.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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